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A multi-step interaction network for multi-class classification based on OCT and OCTA images

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ABSTRACT

OCT and OCTA images are important basis for diagnosing multiple ophthalmic diseases. However, it is a challenge to fuse these two modalities with high redundancy and simultaneous projection of 3D data for multi-class classification. This paper proposes a novel Multi-step Interaction Network (MINet) for projection and feature fusion as a unified framework, where OCT and OCTA images deeply interact for a seven-class classification task of ophthalmic diseases. Firstly, we design a Multi-modal Interaction Projection Module to iterate projection and shallow information interaction for effective feature selection. Secondly, a Feature Redundancy Removal Module compares the feature difference information between the two modalities to eliminate redundancy. Thirdly, the Feature Interaction Fusion Module utilizes the differential modal information from the backbone CNN to perform respective modal attention and achieve interactive fusion. Finally, a classifier module generates multi-class classification results. Experimental results show that our method achieved Accuracy of 0.8690, Precision of 0.6921, Recall of 0.7250, and F1 score of 0.7081 on the OCTA-500 dataset. Comparative experiments with other state-of-the-art methods for OCT image classification, along with ablation experiments, demonstrate the superior performance of MINet.

1. Introduction

Age-related macular degeneration (AMD), central serous chorioretinopathy (CSC) and a series of other retinal diseases are among the major causes of vision loss or impairment. With the aging global population and the increasing number of patients with chronic diseases, the prevalence of eye disorders is showing a continuous upward trend [1]. Timely and accurate diagnosis and treatment of these diseases are crucial for ensuring the visual health of patients.

Optical coherence tomography (OCT) and optical coherence tomography angiography (OCTA) are two important techniques for ophthalmic imaging. Since both have their own unique applications, their multi-modal fusion can complement each other and provide a more comprehensive ophthalmic assessment. OCT is a non-invasive imaging technique primarily used to obtain microstructural images of biological tissues. OCT can provide high-resolution three-dimensional images, making it particularly suitable for observing eye structures such as the retina, optic nerve, and anterior chamber [2], which is of great significance for the diagnosis of ophthalmic diseases. The advantage of OCT is that high-resolution structural imaging provides detailed information about tissue morphology. OCT focuses on evaluating anatomical structures, but it cannot directly observe blood flow. OCTA is built

on the foundation of OCT by imaging blood flow. It utilizes phase changes of moving blood cells at different time points to generate three dimensional(3D) images of the vascular network, which can clearly display the distribution of retinal and subretinal vessels, and assists in the identification of vascular abnormalities, ischemia, or blood flow conditions related to retinal diseases [3]. The advantage of OCTA is that it allows for non-invasive visualization of vascular structures and blood flow conditions, providing insights into vascular health. The limitation of OCTA is that the image quality may be affected by ocular motion or refractive errors, possibly leading to reduced clarity [4].

Different ophthalmic diseases exhibit distinct characteristics in OCT and OCTA images. On OCT images, AMD is characterized by undulating elevations in the Retinal Pigment Epithelium (RPE) layer [5]. OCTA images reveal that macular neovascularization in AMD patients originates within the retina rather than the choroid [6]. Diabetic retinopathy (DR) manifests as microaneurysms in OCT images [7], while OCTA images show intraretinal microvascular abnormalities, which may include vascular rupture, leakage, or blockage [8]. Choroidal neovascularization (CNV) is represented on OCTA as the growth of neovascular tissue beneath the retina and on the RPE [9]. Retinal vein occlusion (RVO) is characterized by structural disturbance of the inner retina in OCT

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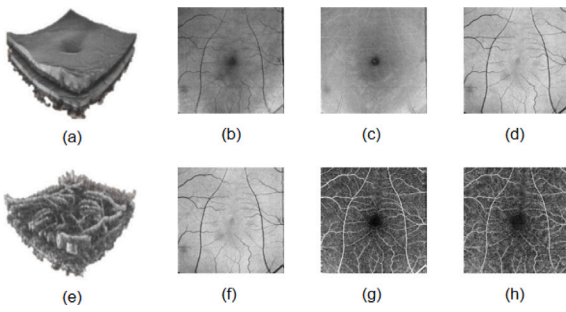


Fig. 1. (a) OCT volume. (b) OCT full projection. (c) OCT average projection between the internal limiting membrane (ILM) layer and the outer plexiform layer (OPL). (d) OCT average projection between OPL and Bruch's membrane (BM). (e) OCTA volume. (f) OCTA full projection. (g) OCTA average projection between ILM and OPL. (h) OCTA average projection between OPL and BM.

images [10] and decreased density of radial peripapillary capillaries (RPC) around the optic disc in OCTA images [11]. Due to the complexity and diversity of their features, multi-class classification for these multiple ophthalmic diseases urgently requires to develop the deep learning model based on fusion of OCT and OCTA.

However, when utilizing deep learning to diagnose ophthalmic diseases through OCT and OCTA images, several main challenges are currently faced. Firstly, owing to 3D data of OCT and OCTA, their large volume results in low computational efficiency. Furthermore, the useful information only accounts for a small portion of the overall dataset, where a significant amount of noisy data can lead to overfitting. So, it is crucial to effectively utilize 3D retinal data. Secondly, because of the large amount of shared feature information between OCT and OCTA images, it is necessary to perform feature redundancy reduction to avoid interference from repeatedly capturing them. This enables the network to focus more on the fusion of discriminative features. Thirdly, due to the interconnectedness of projection, redundancy removal, and fusion, it is also necessary to develop a unified framework for them. Since multi-class classification for multiple ophthalmic diseases requires more refined feature learning, extraction of these diverse features in the multi-modal fusion framework is another challenge.

Based on the high redundancy and strong complementarity of OCT and OCTA images, we design a unified deep interaction framework of 3D projection, redundancy removal, and data fusion, called the Multi-step Interactive Network (MINet). This framework aims to fuse the complementary information between OCT and OCTA images through multi-level interactions for the diagnosis of ophthalmic diseases. The input of MINet includes OCT images, OCTA images, and six types of projection maps contained in OCT and OCTA datasets respectively, as shown in Fig. 1. The final output is the predicted results for different eye diseases of testing samples. MINet mainly consists of the following three parts. Firstly, the Multi-modal Interactive Projection Module (MIPM) is utilized to convert 3D image data into 2D projection images by iterating multi-scale projection and shallow feature interaction. Secondly, the Feature Redundancy Removal Module (FRRM) can remove redundancy and pay attention to the differences of feature information between the two modalities. Thirdly, the Feature Interaction Fusion Module (FIFM) enhances the feature information of the two modalities from the backbone Convolutional Neural Network (CNN) by interacting with differential modal information, and finally the classifier module generates classification results. We evaluated the performance of MINet on the OCTA-500 dataset, conducting ablation studies and comparative experiments against other state-of-the-art methods for OCT image classification.

In summary, this paper has the following contributions: (1) We propose a novel framework, called Multi-step Interaction Network (MINet), to project and fuse OCT and OCTA 3D images for the multi-class classification of multiple ophthalmic diseases. (2) In the 3D projection

learning part, we introduce a Multi-modal Interactive Projection Module (MIPM). It iteratively operates multi-scale projection and shallow information interaction during data compression, so as to achieve effective feature selection in the process of projection. (3) In the redundancy removal section, we designed the Feature Redundancy Removal Module (FRRM) to eliminate redundant feature information. It focuses more on discriminative features to improve classification accuracy and reduce the risk of overfitting. (4) In the information fusion section, we propose the Feature Interaction Fusion Module (FIFM), where the interactive attention mechanism utilizing differential modal information is designed for complementary enhancement.

The overall workflow of this paper is as follows: Section 2 reviews the relevant works, Section 3 presents our multi-step interaction network for multi-class classification of ophthalmic diseases, and Section 4 details various experimental designs and the results. Finally, Section 5 summarizes the conclusions of the proposed work.

2. Related works

2.1. CNNs for ophthalmic disease diagnosis

In recent years, with the development of CNN applications in the field of medical image analysis, the technology of computer-aided diagnosis for fundus diseases has made remarkable progress. Deep learning has been widely used in the diagnosis of various ophthalmic diseases, including glaucoma, DR, AMD, cataract and diabetic macular edema (DME) [12].

AlexNet [13], VGG [14], GoogLeNet [15] and ResNet [16] are widely used classification backbone networks for the diagnosis of ophthalmic diseases. DR is one of the most common lesions for deep learning diagnosing eye diseases. Based on AlexNet and VGG, David et al. [17] proposed a system that can automatically detect DR, called IDx-DR X2.1. Gargeya and Leng [18] used a CNN containing five residual blocks to extract image features for DR diagnosis. In addition to the binary diagnosis of DR, based on the International Clinical Diabetic Retinopathy Scale (ICDRS) [19], the extent of DR is categorized into five distinct levels: 0 representing no apparent DR, 1 representing mild DR, 2 representing moderate DR, 3 representing severe DR, and 4 representing proliferative DR. Yang et al. [20] proposed a two-stage Deep Convolutional Neural Networks (DCNN) which can diagnose DR and grade severity. The first stage is an extracting local feature for lesion detection, and the second stage is a global network for grading, which completes multi-class classification tasks. Deep learning for AMD diagnosis has also achieved good results. Burlina et al. [21] assessed the severity of AMD and predicted the risk over the next five years. With the continuous improvement of data, CNNs have gradually developed to diagnose a variety of diseases at the same time. The difficulty of this work is that the recognition of different ophthalmic diseases may affect each other. Different correlation measurements can also be used for image classification prediction [22,23]. Li et al. [24] designed a Dense Correlation Network (DCNet) for the diagnosis of eight different diseases through fundus photographs. The simultaneous diagnosis of multiple diseases with CNN can ensure the comprehensiveness of diagnosis and improve diagnostic efficiency. However, many retinal diseases, such as DR, dry macular degeneration and glaucoma, share similar early symptoms, especially in imaging manifestations, which may lead to misdiagnosis or missed diagnosis. Currently, there is relatively little research on multi-class classification algorithms for multiple ophthalmic diseases, which highlights the urgent need for a novel algorithm to multiple ophthalmic diseases.

2.2. OCT images and OCTA images for ophthalmic disease recognition

At present, the screening of eye diseases is primarily divided into fundus image recognition and OCT image recognition. Because the collection of fundus images is more convenient, most CNNs use fundus images to diagnose ocular diseases. However, compared with OCT and OCTA images, fundus images have a lower resolution, which makes it difficult to display the small structural changes of the retina clearly. Additionally, fundus images are two-dimensional, which prevents them from providing depth information of the various retinal layers, thus limiting the ability to identify layered diseases [25]. Therefore, OCT and OCTA have significant advantages for ophthalmic disease recognition.

Since OCT and OCTA data are composed of 3D data, their large data volume leads to low computational efficiency. Additionally, the key information related to disease diagnosis constitutes only a small part of the overall data, which can easily lead to overfitting. It is a significant challenge to effectively utilize 3D retinal data. Thakoor et al. [26] proposed a hybrid 3D-2D CNNs for distinguishing whether AMD patients are neovascularized. Li et al. [27] introduced a 3D to 2D segmentation network, image projection network (IPN), designed for the segmentation task of OCTA images. They built a projection learning module (PLM) that aggregates valid information from 3D data into a 2D projection plane. More importantly, these methods overlooked the differences between the two modalities of OCT and OCTA, failing to fully leverage the complementary information for ophthalmic disease recognition.

2.3. Attention mechanism

Features associated with ophthalmic diseases are usually concentrated in specific areas of the image, such as the retina and optic nerve. In OCT and OCTA images, there is a significant amount of noise and irrelevant information. Therefore, attention techniques in deep learning are particularly important to assist in automatically identifying and focusing on key feature areas. Hu et al. [28] proposed Squeeze-and-Excitation Networks (SENet), which models the interdependencies between channels to learn the importance of different channels. SENet can adaptively weight them and recalibrate the feature responses of the channels, so as to enhance the influence of useful feature channels and suppress those of useless feature channels.

OCT and OCTA images contain information at multiple scales. An appropriate attention mechanism can adaptively extract features from different scales, thereby enhancing the sensitivity to subtle changes in diseases. Different sizes of receptive fields, namely convolutional kernels, have various effects. Li et al. [29] proposed Selective Kernel Networks (SKNet), which adaptively selects more important convolutional kernel sizes, effectively integrating multi-scale feature information.

However, the features of many ophthalmic diseases usually require a comprehensive consideration of the overall contextual information in the image, not limited to a specific local area, where attention mechanisms can effectively capture long-range dependencies. In the paper of Non-local Neural Networks, Wang et al. [30] proposed a non-local attention mechanism. Compared to attention mechanisms like SENet that stack convolutions, the non-local attention mechanism directly computes the relationships between two points to quickly capture long-range dependencies. Based on non-local neural networks, Ou et al. [31] designed a feature interaction module for the interaction between bilateral fundus images.

This paper aims to present a multi-level feature interaction framework based on the non-local attention. It captures the global dependencies between feature information for multi-modal fusion, thereby improving the classification ability of the network.

3. Methods

3.1. The overall framework

The framework of our proposed Multi-step Interaction Network (MINet) is illustrated in Fig. 2 and mainly consists of three parts: the 3D interaction projection part, the feature redundancy removal part, and the feature interaction fusion part. In the 3D interaction projection module, the Multi-modal Interactive Projection Module (MIPM) iteratively performs multi-scale projection and shallow interaction three times to compress the volumetric data of OCT and OCTA into 2D projections. Specifically, shallow information interaction operation is performed to fuse the feature information of OCT and OCTA images after each projection, including image features and semantic features, thereby achieving effective feature selection during the projection process. In the feature redundancy removal part, we compare the differences in feature information between the two modalities to eliminate redundant information. The differential feature information obtained through redundancy removal is enhanced using an attention mechanism, based on the original feature information of each modality, to strengthen representation. In the feature interaction fusion part, we extract the feature information from both modalities through the backbone CNN. Subsequently, we once again obtain the differential modal feature for multi-modal interaction fusion, where the interdependence between the single-modal information and the differential information can be learned through attention mechanism. These feature weights are adjusted based on importance to achieve complementary enhancement. Finally, the enhanced feature information is fed into the classifier to complete the multi-class classification task of ophthalmic diseases.

3.1.1. Multi-modal Interactive Projection Module (MIPM)

Hou et al. [32] proposed a Kernel Adaptive Projection Module (KAPM) that achieves kernel adaptive multi-scale convolutional projection of 3D image information for segmentation. It does not efficiently utilize the complementarity of information and involves a large computational load. In order to achieve more effective projection, this paper proposes a layered unidirectional convolutional projection method based on original unidirectional convolution, which can avoid the loss of hierarchical feature information as much as possible during the projection process, as shown in Fig. 3(a). A multi-scale projection module is also set to capture different resolution information and obtain multi-scale features. Furthermore, to enhance modality interaction, shallow information interaction operations are integrated into the projection process. Compared to KAPM, MIPM iteratively combines projection and shallow information interaction operations into a linear structure, achieving effective feature selection during the projection process. MIPM consists of three mipm units. Fig. 3(b) shows the workflow of a mipm unit.

We divide the 3D volume data of size (L, H, W, C_{in}) into n equal parts along the L -dimension direction and perform unidirectional convolutional projection on each part along the L -dimension. Then, we obtain n sets of 3D volumetric data, each of size $(1, H, W, C_{out})$, and we concatenate these n compressed data. After the layered unidirectional convolutional projection, we compress the size of the 3D volume data from (L, H, W, C_{in}) to (n, H, W, C_{out}) , where $n < L$. Since OCT and OCTA images contain rich depth and structural information, we can capture feature information at different depth levels through layered projection, thereby preserving important anatomical structures and pathological characteristics. The change of size is shown in Table 1(a). After three projection steps, the 3D volume data is compressed into 2D.

We also design convolution of different kernel-sizes to capture information at various resolutions, extracting features at different scales to obtain rich global and local information, as OCT and OCTA images contain information at multiple scales. In this paper, we project 3D volume data using 3D convolution with $L/n \times 3 \times 3$ and $L/n \times 5 \times 5$ kernel-size respectively. The compressed data information is

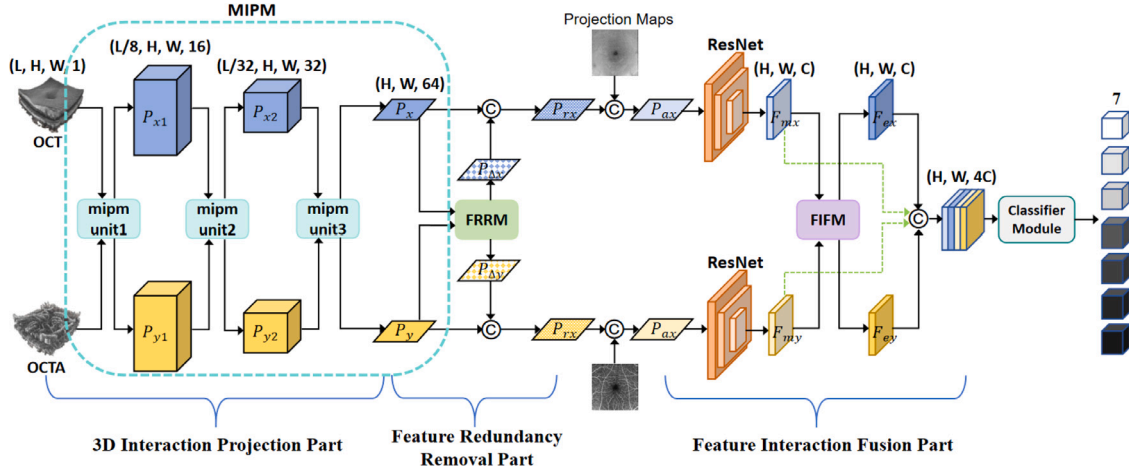


Fig. 2. The overall framework of MINet. “C” is concatenation.

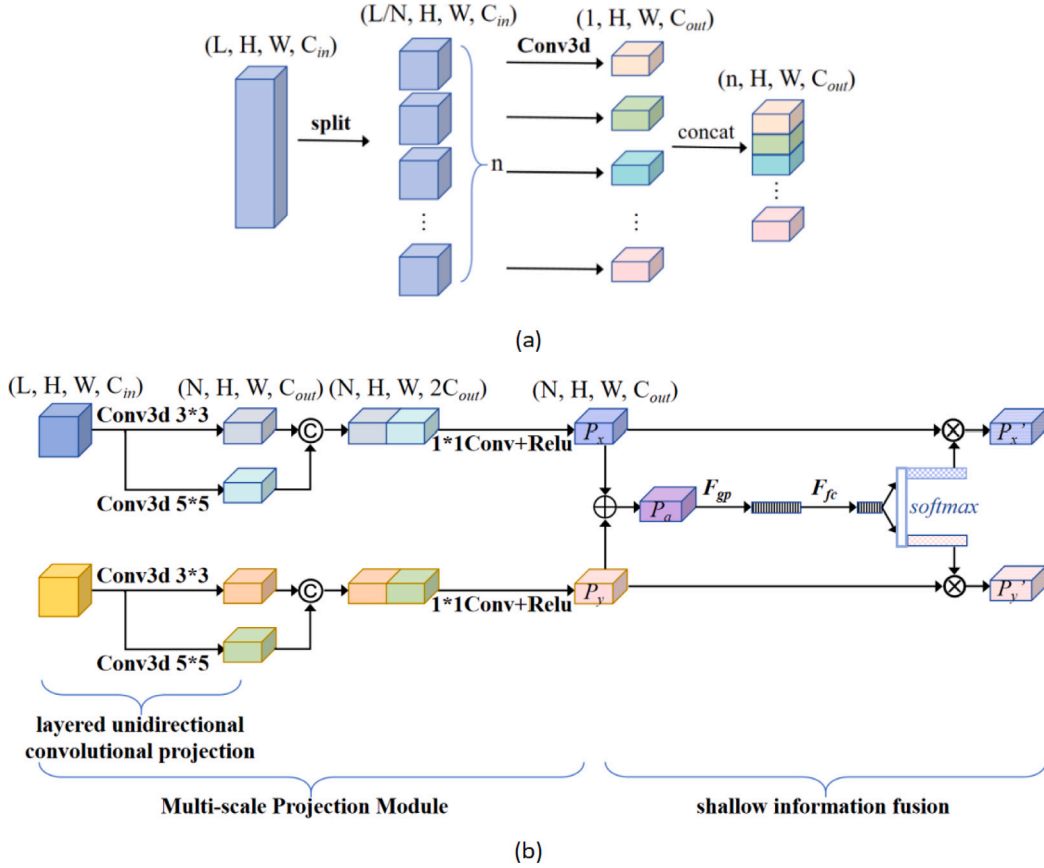


Fig. 3. (a) The structure of the layered unidirectional convolutional projection. (b) The structure of mipm unit. “ F_{gp} ” is the global average pooling function, “ F_{fc} ” is the fully connected layer.

concatenated to a volume data of size $(N, H, W, 2C_{out})$, and after a 1×1 convolution and a ReLU function, the size becomes (N, H, W, C_{out}) . The sizes of the convolutional kernels at the projection process and the changes of data dimensions are shown in Table 1(a).

MIPM makes the two modalities interact for multi-level, in order to better fuse the information of OCT and OCTA. After each compression of the 3D volumetric data, we perform interactive operations of shallow information, so that the image information and semantic information from both modalities interacts. As the degree of volumetric data compression increases, the quantity and depth of the interactive semantic information gradually increases. As shown in Fig. 3(b), the projection

information P_x and P_y are added together, and then pass through a pooling layer and a fully connected layer for the information interaction from the two modalities. The pooling layer is a global average pooling function, and the fully connected layer contains two 1×1 convolutions, with a ReLU function inserted after the first convolution operation. Then, a Softmax layer is applied for selection, and the results are multiplied with P_x and P_y respectively to obtain the projection images P'_x and P'_y after interactive guidance.

After comprehensive experimental evaluation, we chose to compose the MIPM using three mipm units. The MIPM parameters are shown

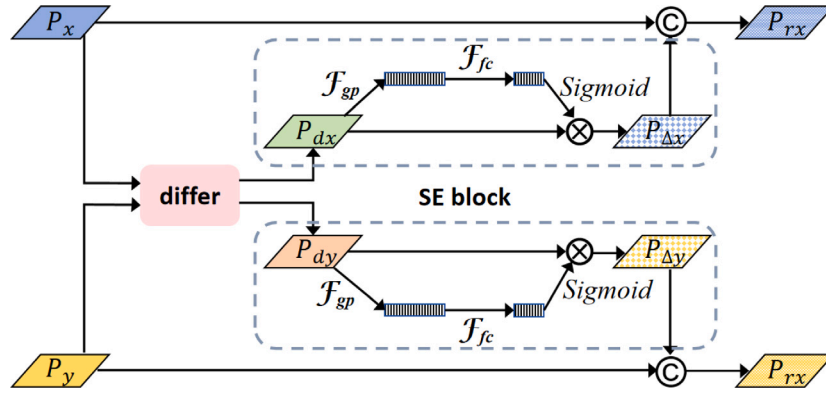


Fig. 4. The structure of FRRM.

Table 1

The parameters of MIPM.

Stage	Input size	Operation	Output size
mipm unit1	(128,W,H,1)	$MPconv(k = (8,3,3), s = (8,1,1))$	(16,W,H,16)
		$MPconv(k = (8,5,5), s = (8,1,1))$	
		STF	
mipm unit2	(16,W,H,16)	$MPconv(k = (4,3,3), s = (4,1,1))$	(4,W,H,32)
		$MPconv(k = (4,5,5), s = (4,1,1))$	
		STF	
mipm unit3	(4,W,H,64)	$MPconv(k = (4,3,3), s = (4,1,1))$	(W,H,64)
		$MPconv(k = (4,5,5), s = (4,1,1))$	
		STF	

in Table 1. Where “ $MPconv$ ” refers to the multi-scale projection convolution, “ k ” is the kernel size, “ s ” is the stride, and “STF” is the shallow information fusion. STF consists of an average pooling layer, a fully connected layer, and a Softmax function. After the MIPM, the 3D information is projected into 2D. The mipm unit3 gets the volume data of size (1, W, H, 64), and the squeeze function is used to remove the dimension of size 1, resulting in a final output size of (W, H, 64).

$$P_x = Relu(Conv1(Concat(Conv3(P_{OCT}), Conv5(P_{OCT}))), \quad (1)$$

$$P_y = Relu(Conv1(Concat(Conv3(P_{OCTA}), Conv5(P_{OCTA}))), \quad (2)$$

$$P'_x = Softmax(Conv(Relu(Conv(AvgPool3d(P_x + P_y))))[0] \times P_x, \quad (3)$$

$$P'_y = Softmax(Conv(Relu(Conv(AvgPool3d(P_x + P_y))))[1] \times P_y. \quad (4)$$

3.1.2. Feature Redundancy Removal Module (FRRM)

Since OCT and OCTA images contain rich feature information respectively, it is important to extract the differential information to strength their respective features. However, OCT and OCTA share a significant amount of common feature information, it is essential to perform removal operations of feature redundancy to prevent differential information from being overwhelmed. This allows the network to focus more on distinctive features, thereby improving efficiency and reducing the risk of overfitting.

In order to avoid redundant features introduced by directly concatenating or adding multi-modal data information, Zhou Kailai et al. [33] proposed an Illumination Aware Feature Alignment Module (IAFA). However, this approach focuses on enhancing one modality with the other modality, which may cause a modality imbalance problem. Since the two modalities of OCT and OCTA have balanced characteristics in clinical diagnostic applications, the diagnosis of conditions such as AMD, DR, and RVO requires not only observing the morphological and structural traits of retinal cells based on OCT images, but also observing the blood vessel density and shape based on OCTA images. Inspired by above point, we propose a modality symmetric balanced Feature Redundancy Removal Module (FRRM). The main idea of this method is to compare the difference of the feature information between the

two modalities, remove the redundant common information, and then mutually enhance the de-redundant differential feature information with the original feature information. This symmetric approach fully exploits the inherent complementarity between modality features to avoid imbalanced modal enhancement.

As shown in Fig. 4, P_x and P_y represent the 2D data information output from the last MIPM. By comparing the differences between them, P_{dx} and P_{dy} are obtained. We utilize the Squeeze-and-Excitation (SE) block of channel attention to learn the differential feature weights, resulting in the differential information with the redundant information removed, $P_{\Delta x}$ and $P_{\Delta y}$. The SE block consists of an average pooling layer, a fully connected layer, and a Sigmoid function. The fully connected layer includes two 1×1 convolutions, with a ReLU function inserted between them. Finally, concatenate P_x and $P_{\Delta x}$ to obtain P_{rx} , and concatenate P_y and $P_{\Delta y}$ to obtain P_{ry} .

$$P_{dx} = P_x - P_y, \quad (5)$$

$$P_{dy} = P_y - P_x, \quad (6)$$

$$P_{\Delta x} = Sigmoid(Conv(Relu(Conv(AvgPool(P_{dx})))))) \times P_{dx}, \quad (7)$$

$$P_{\Delta y} = Sigmoid(Conv(Relu(Conv(AvgPool(P_{dy})))))) \times P_{dy}, \quad (8)$$

$$P_{rx} = Concat(P_x, P_{\Delta x}), \quad (9)$$

$$P_{ry} = Concat(P_y, P_{\Delta y}). \quad (10)$$

3.1.3. Feature Interaction Fusion Module (FIFM)

To make comprehensive use of both global and local features in OCT and OCTA for recognition of multiple ophthalmic diseases, this paper proposes a multi-modal interactive attention based on the self-attention, named FIFM. This attention mechanism highlights the specific features of each modality in the global information, facilitating the utilization of complementary information for further enhancement. Because the traditional direct connection scheme to capture cross-modal complementary information is ambiguous and tends to mix repeated information. FIFM achieves interaction between the two modalities by the differential modality information. FIFM takes the differential modality information as local information and queries the global information of each modality separately, thus avoiding the redundancy caused by concatenation way. The novel attention mechanism highlights the specific features of each modality by facilitating reuse of differential modality.

As shown in Fig. 5, the input of FIFM, namely F_{mx} and F_{my} , are the feature information of the two modalities extracted by the backbone CNN. By comparing F_{mx} and F_{my} , we derive the differential modality information $F_{\Delta x}$ and $F_{\Delta y}$ as local information. Then, we convolve $F_{\Delta x}$ and $F_{\Delta y}$ to convert them to Q (Query), Q_x and Q_y . The global information F_{mx} and F_{my} are also convolved to K (Key), K_x and K_y , as well as V (Value), V_x and V_y . The calculation formulas of Q, K and V are as follows:

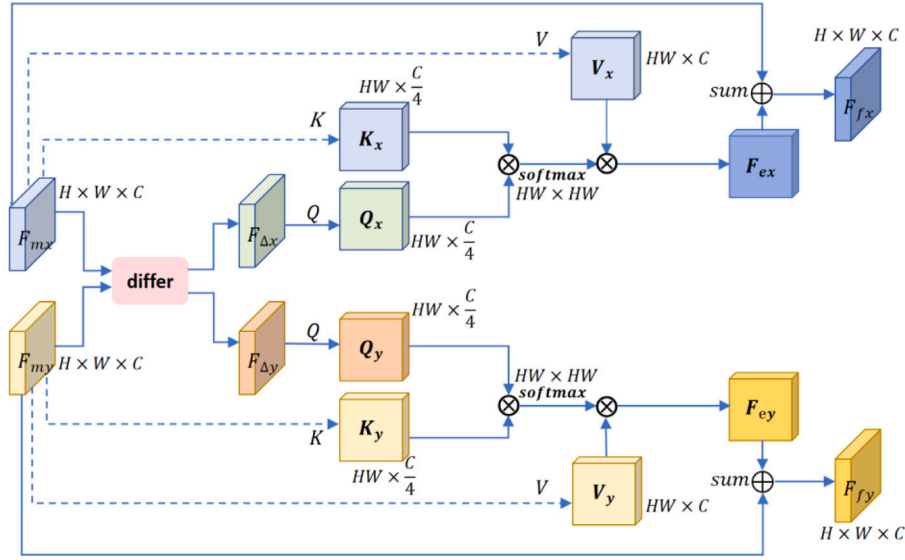


Fig. 5. The structure of FIFM.

$$F_{\Delta x} = F_{mx} - F_{my}, \quad (11)$$

$$F_{\Delta y} = F_{my} - F_{mx}, \quad (12)$$

$$K_x = \text{Conv}(F_{mx}; \theta_{qx}), \quad (13)$$

$$K_y = \text{Conv}(F_{my}; \theta_{qy}), \quad (14)$$

$$Q_x = \text{Conv}(F_{\Delta x}; \theta_{kx}), \quad (15)$$

$$Q_y = \text{Conv}(F_{\Delta y}; \theta_{ky}), \quad (16)$$

$$V_x = \text{Conv}(F_{mx}; \theta_{vx}), \quad (17)$$

$$V_y = \text{Conv}(F_{my}; \theta_{vy}), \quad (18)$$

where Conv is 1×1 convolution, and θ with different subscripts represent relevant parameters.

In order to calculate the correlation between the local differential information and global information for OCT, we aggregate the local differential information into the global information, calculate their inner product, and then normalize the inner product with a Softmax function for the correlation weights. The method of calculating the correlation weights for OCTA is the same. The correlation weights are multiplied with the global information to obtain the enhanced feature information F_{ex} and F_{ey} . Respectively add F_{ex} and F_{ey} to F_{mx} and F_{my} , resulting in the enhanced feature maps F_{fx} and F_{fy} .

$$F_{ex} = \text{Softmax}(Q_x^T K_x) \times V_x, \quad (19)$$

$$F_{ey} = \text{Softmax}(Q_y^T K_y) \times V_y. \quad (20)$$

Finally, we design a residual structure, that respectively add F_{mx} and F_{my} to F_{ex} and F_{ey} , resulting in the enhanced feature graphs F_{fx} and F_{fy} , which effectively alleviate gradient disappearance in the interactive attention process of the two modalities.

$$F_{fx} = F_{mx} + F_{ex}, \quad (21)$$

$$F_{fy} = F_{my} + F_{ey}. \quad (22)$$

3.1.4. The classifier module

As shown in Fig. 1, before being input into the classifier module, the enhanced feature maps F_{fx} and F_{fy} from FIFM are concatenated with the feature maps F_{mx} and F_{my} extracted from the backbone CNN. The classifier module consists of three fully connected layers, with a ReLU activation function between every two layers. The fully connected layers gradually reduce the feature dimension, ultimately compressing the feature dimension to 7, which corresponds to the classification

categories. The classification results are then compared with the actual labels to calculate the network loss.

3.2. Loss function

The commonly used loss function for classification tasks is the Cross Entropy Loss, but it is not suitable for multi-class classification task. According to the existing research [34], this paper adopts the Binary Cross Entropy Loss to calculate the network loss:

$$BCELoss(y, \hat{y}) = -\frac{1}{N} \sum_{i=0}^N (y \times \log(\hat{y}) + (1 - y) \times \log(1 - \hat{y})), \quad (23)$$

where N represents the number of categories, y indicates the actual label, and $\hat{y}$ indicates the predicted label.

However, since the dataset used in this paper belongs to long tail dataset, that has an imbalanced distribution of samples across different labels. In order to avoid insufficient representation learned from too little tail data, we assign the loss weights according to the proportion of each category. The loss function is calculated as follows:

$$BCELoss(y, \hat{y}) = -\frac{1}{N} \sum_{i=0}^N (y \times \log(\hat{y}) + (1 - y) \times \log(1 - \hat{y})) \times W_i, \quad (24)$$

where W_i indicates the loss weight of each category.

4. Experiments

This section first introduces the dataset used in this paper, the evaluation metrics, and main details. Then, we give ablation experiments and comparative experiments. The ablation experiments investigated the effects of different layer structure projections, the impact of different backbone CNNs, and the influence of different modules in MINet. The comparative experiments included comparisons with state-of-the-art methods and classical methods.

4.1. Datasets

In this paper, we chose the OCTA-500 dataset [35] to evaluate MINet. The OCTA-500 dataset provides 3D volumetric data in both OCT and OCTA modalities, along with six types of 2D projection images. The dataset contains a total of 500 samples across seven label categories. These label categories include NORMAL, age-related macular degeneration (AMD), diabetic retinopathy (DR), choroidal neovascularization (CNV), retinal vein obstruction (RVO), central serous choroidal retinopathy (CSC), and other diseases (OTHERS). There are 251 NORMAL samples, 49 AMD samples, 64 DR Samples, 16 CNV

Table 2
The category distribution of dataset.

	NORMAL	AMD	DR	CNV	RVO	CSC	OTHERS	Total
Train	152	34	37	9	6	7	55	300
Validation	22	1	9	2	1	1	14	50
Test	77	14	18	5	3	6	27	150
Total	251	49	64	16	10	14	96	500

Table 3
The classification effect of MINet for various categories.

	Acc	Pre	Rec	F1
NORMAL	0.8400	0.8052	0.8732	0.8378
AMD	0.8667	0.5000	0.3500	0.4118
DR	0.9533	0.8333	0.7894	0.8108
CNV	0.9733	0.4000	0.6667	0.5000
RVO	0.9800	0.3333	0.5000	0.4000
CSC	0.9667	0.6667	0.5714	0.6154
OTHERS	0.8067	0.4815	0.4643	0.4727
Total	0.8620	0.6933	0.7181	0.7055

samples, 10 RVO samples, 14 CSC samples, and 96 OTHERS samples. The category distribution of training set, validation set and test set is shown in Table 2.

Due to the limited number of samples in the tail categories, we adopted the resampling method for data enhancement by randomly sampling with replacement in proportion, expanding the training set to 400 samples.

4.2. Evaluation metrics

To evaluate the classification performance of our network, this paper used Accuracy, Precision, Recall, and F1 score as evaluation metrics to assess the classification results for each category as well as overall performance. We denote the Accuracy, Precision, Recall, and F1 score for each category as Acc_i , Pre_i , Rec_i , and $F1_i$ (where $i \in [1,7]$), while Acc , Pre , Rec , and $F1$ represent the overall Accuracy, Precision, Recall, and F1 score.

$$Acc_i = \frac{TP_i + TN_i}{TP_i + FP_i + FN_i + TN_i}, \quad (25)$$

$$Pre_i = \frac{TP_i}{TP_i + FP_i}, \quad (26)$$

$$Rec_i = \frac{TP_i}{TP_i + FN_i}, \quad (27)$$

$$F1_i = \frac{2 \times Pre_i \times Rec_i}{Pre_i + Rec_i}. \quad (28)$$

Due to the sample distribution in the test set was imbalanced, the calculations for Acc , Pre , and Rec were performed using a weighted average approach, specifically:

$$Acc = Acc_i \times W_i, \quad (29)$$

$$Pre = Pre_i \times W_i, \quad (30)$$

$$Rec = Rec_i \times W_i, \quad (31)$$

$$F1 = \frac{2 \times Pre \times Rec}{Pre + Rec}, \quad (32)$$

where W_i represents the weight of each category in the test set.

4.3. Implementation details

We built our network with the PyTorch framework, and all experiments were implemented on the GeForce RTX 3090 Ti GPU. We employed the Adaptive Moment Estimation (Adam) optimizer [36] to train the network, setting the first coefficient to 0.9, the second coefficient to 0.99, the initial learning rate to 0.003, batch size to 4, and experiment iteration number to 200 epochs.

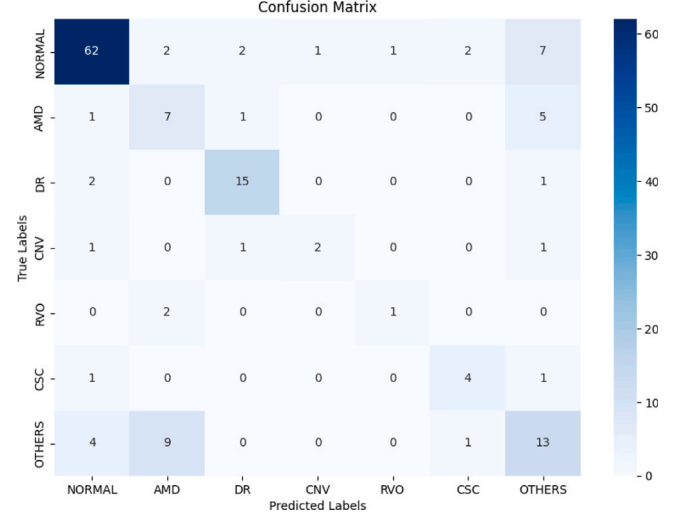


Fig. 6. Confusion matrix of each category.

4.4. The classification performance of MINet

MINet is the first model for the seven-class classification and demonstrates excellent performance on OCT and OCTA image datasets, especially excelling with Rec and $F1$, as shown in Table 3.

However, some categories have lower precision or recall. To further investigate the classification status of each category, we provide a confusion matrix reflecting the actual classification results for each category, as shown in Fig. 6.

According to the experimental results, both AMD and OTHERS have relatively low precision and recall. This is because the main manifestation of AMD in OCT images is macular lesions [37], which may be confused with samples of other macular diseases in the OTHERS category. Additionally, the precision of RVO is low, that is due to the similarity in the manifestation of fluid accumulation and changes in the macular area in OCT images, which can easily be confused with the appearance of AMD. According to related studies [38,39], while OCT and OCTA images have significant auxiliary diagnostic effects, they still cannot replace the role of angiography in the diagnosis of AMD and RVO. The gold standard for detecting AMD and RVO is still the combined imaging examination of Fluorescein Fundus Angiography (FFA) and Indocyanine Green Angiography (ICGA).

4.5. Ablation experiments

4.5.1. The role of different layer structures in OCT and OCTA images

To further explore the roles of different layer structures in OCT and OCTA images, we designed ablation experiments for different projection images. We show the average results of 10 runs in Table 4. The first experiment was conducted without the average projection between the ILM and the OPL, and the second experiment was conducted without the average projection between the OPL and the BM. The results showed that both the ILM-OPL structure and the OPL-BM structure play important roles in disease diagnosis, and the ILM-OPL structure projection played a significant role.

Table 4

The impact of different layer structures in OCT and OCTA images on classification performance.

	Acc	Pre	Rec	F1
MINet without ILM-OPL	0.8563 $\pm$ 0.0164	0.6566 $\pm$ 0.0129	0.6513 $\pm$ 0.0145	0.6519 $\pm$ 0.0153
MINet without OPL-BM	0.8528 $\pm$ 0.0166	0.6783 $\pm$ 0.0158	0.6652 $\pm$ 0.0114	0.6781 $\pm$ 0.0137
Ours	0.8685 $\pm$ 0.0129	0.6899 $\pm$ 0.0304	0.7241 $\pm$ 0.0172	0.7066 $\pm$ 0.0229

Table 5

The impact of the backbone CNN on classification performance.

	Acc	Pre	Rec	F1
MINet-VGG16	0.8254 $\pm$ 0.0173	0.6262 $\pm$ 0.0248	0.6434 $\pm$ 0.0164	0.6385 $\pm$ 0.0186
MINet-AlexNet	0.8162 $\pm$ 0.0147	0.6205 $\pm$ 0.0137	0.6360 $\pm$ 0.0153	0.6329 $\pm$ 0.0134
MINet-DenseNet	0.8469 $\pm$ 0.0127	0.6753 $\pm$ 0.0162	0.6504 $\pm$ 0.0195	0.6698 $\pm$ 0.0160
MINet-ResNet-18	0.8514 $\pm$ 0.0104	0.6756 $\pm$ 0.0157	0.6589 $\pm$ 0.0121	0.6681 $\pm$ 0.0138
MINet-ResNet-34	0.8685 $\pm$ 0.0129	0.6899 $\pm$ 0.0304	0.7241 $\pm$ 0.0170	0.7066 $\pm$ 0.0229
MINet-ResNet-50	0.8394 $\pm$ 0.0238	0.6648 $\pm$ 0.0175	0.6407 $\pm$ 0.0228	0.6528 $\pm$ 0.0202

Table 6

The impact of different modules.

	Acc	Pre	Rec	F1
MINet without multi-scale projection module	0.8590 $\pm$ 0.0132	0.6813 $\pm$ 0.0146	0.6865 $\pm$ 0.0198	0.6836 $\pm$ 0.0171
MINet without shallow information fusion	0.8468 $\pm$ 0.0154	0.6687 $\pm$ 0.0124	0.6958 $\pm$ 0.0169	0.6895 $\pm$ 0.0118
MINet without FRRM	0.8526 $\pm$ 0.0261	0.6823 $\pm$ 0.0236	0.6875 $\pm$ 0.0152	0.6854 $\pm$ 0.0165
MINet without FIFM	0.8396 $\pm$ 0.0159	0.6495 $\pm$ 0.0169	0.6598 $\pm$ 0.0189	0.6552 $\pm$ 0.0185
MINet with concat-FIFM	0.8336 $\pm$ 0.0148	0.6758 $\pm$ 0.0123	0.6726 $\pm$ 0.0164	0.6754 $\pm$ 0.0155
MINet with differ-concat-FIFM	0.8353 $\pm$ 0.0163	0.6651 $\pm$ 0.0138	0.6523 $\pm$ 0.0134	0.6551 $\pm$ 0.0135
Ours	0.8685 $\pm$ 0.0129	0.6899 $\pm$ 0.0304	0.7241 $\pm$ 0.0172	0.7066 $\pm$ 0.0229

Table 7

Comparative calculation parameters of network with other classification networks.

	FLOPs ($\times 10^9$)	Params ($\times 10^6$)
ResNet	37.0	31.2
SE-ResNet	52.9	59.6
CBAM-ResNet	92.4	82.5
AOCT-Net	2345.4	498.8
TCAM-ResNet	2639.5	521.4
Ours	1831.9	479.5

4.5.2. The effect of backbone CNN

We designed an ablation experiment to explore the impact of the backbone CNN and its depth on classification performance. As the average results of 10 runs shown in Table 5, when the backbone CNN is ResNet-34, MINet achieved the best results. So, we chose ResNet-34 as the backbone CNN.

4.5.3. Module ablation experiment

Finally, in order to test the effect of each module on classification performance of the network, we designed the module ablation experiments.

To test the effect of MIPM on the classification results, we designed two ablation experiments. One was to remove the multi-scale projection module and replace the projection with a single-size convolution, resulting in F1 being 2.99% lower than MINet. The other was to remove of shallow information fusion, resulting in F1 being 1.94% lower than MINet.

To test the effect of FRRM, we designed an ablation experiment to remove FRRM, resulting in F1 being 1.98% lower than MINet.

To test the effect of FIFM, we designed three sets of ablation experiments. The first was to remove FIFM, resulting in F1 being 5.66% lower than MINet. The second was to replace the differential modality information interaction in FIFM with a concatenation interaction, as shown in the structure of Fig. 7(a). The features F_{mx} and F_{my} extracted by the backbone CNN were concatenated together to form the global information F_{mg} . The feature information of two modalities, F_{mx} and F_{my} , were taken as the local information. The concat-FIFM attention mechanism queried the local information of each modality within the global information of the concatenated dual modalities, to obtain

the attention weights of the OCT and OCTA features. This approach resulted in an F1 score that is 3.36% lower than that of MINet. The third was to replace the single modality information in FIFM with concatenated information, as shown in the structure of Fig. 7(b). This combined the differential modality information interaction with concatenation. The concatenated modality information was taken as global information F_{mg} , and the differential modality information was taken as local information F_{ml} . The differ-concat-FIFM attention mechanism queried the local information of differential modality features within the global information of the concatenated dual modalities, to obtain the attention weights of the concatenated modality features. And its F1 score was 5.40% lower than MINet. The above three experiments demonstrate that the FIFM with differential modality information interaction can effectively avoid global information redundancy caused by concatenation interaction, and realize the interaction enhancement of OCT and OCTA modality information.

The average results of 10 runs for the module ablation experiments are shown in Table 6.

4.6. Comparative experiments of minet and other classification networks

To validate the performance of our proposed network, we conducted comparative experiments with state-of-the-art methods of OCT image classification such as TCAM-ResNet [40], and AOCT-Net [41], as well as baseline classification methods like ResNet, SE-ResNet, and CBAM-ResNet. In these experiments, ResNet, SE-ResNet, CBAM-ResNet [42] used the 2D projection images provided by the OCTA-500 dataset.

Table 7 lists the calculation parameters of these models. Since ResNet, SE-ResNet and CBAM-ResNet only use the two-dimensional projection images, their complexity is significantly lower. In contrast, the TCAM-ResNet and AOCT-Net models have higher complexity because they do not reduce the dimensionality of the 3D images through projection. Our model achieves best performance in 3D image classification models.

Our method achieved the best performance and experimental results are shown in Table 8. The F1 scores of ResNet, SE-ResNet, and CBAM-ResNet were lower than those of other networks. Because the 2D projection images provided by dataset lack hierarchical structure information compared to the 3D OCT images, resulting in inferior

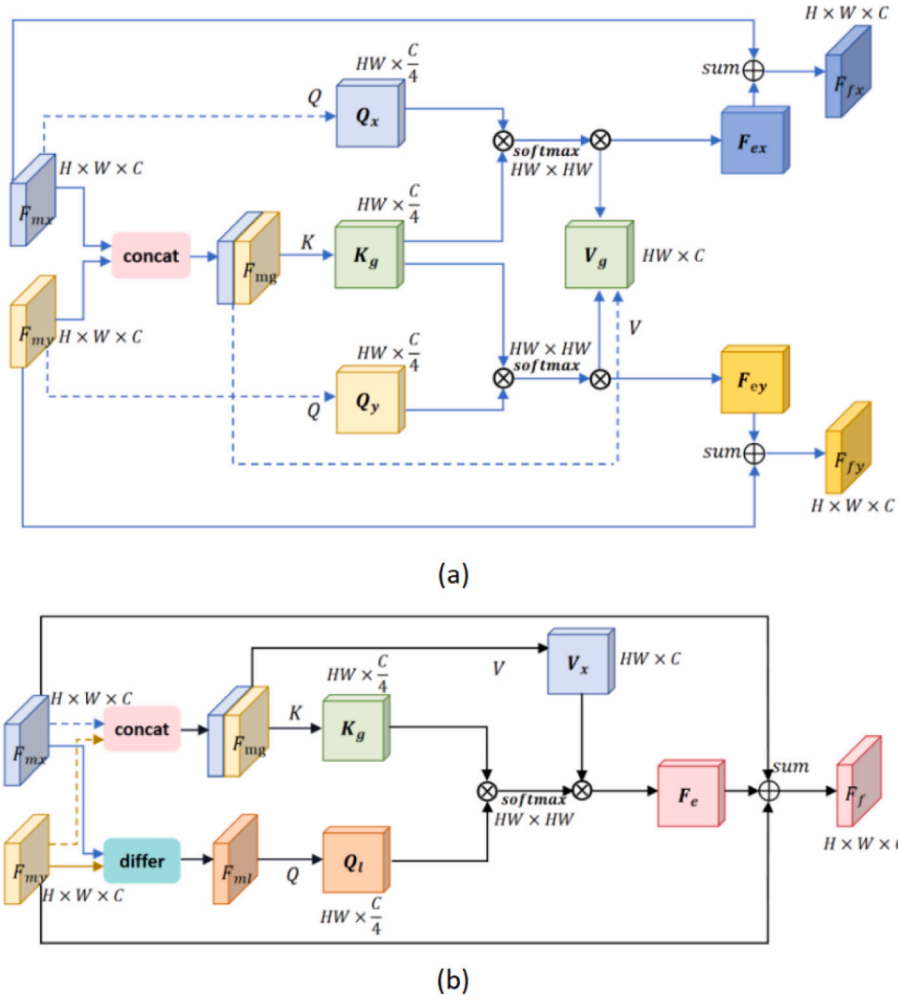


Fig. 7. (a): The structure of concat-FIFM. (b): The structure of differ-concat-FIFM.

Table 8
Comparative experiments with other classification networks.

	Acc	Pre	Rec	F1
ResNet	0.8362 ± 0.0131	0.6492 ± 0.0109	0.6263 ± 0.0163	0.6375 ± 0.0137
SE-ResNet	0.8391 ± 0.0093	0.6701 ± 0.0209	0.6333 ± 0.0171	0.6509 ± 0.0183
CBAM-ResNet	0.8443 ± 0.0164	0.6612 ± 0.0231	0.6734 ± 0.0129	0.6659 ± 0.0126
AOCT-Net	0.8437 ± 0.0135	0.6592 ± 0.0113	0.6472 ± 0.0164	0.6531 ± 0.0132
TCAM-ResNet	0.8494 ± 0.0122	0.6427 ± 0.0100	0.6862 ± 0.0323	0.6619 ± 0.0194
Ours	0.8685 ± 0.0129	0.6899 ± 0.0304	0.7241 ± 0.0172	0.7066 ± 0.0229

classification performance for the 2D networks. Among them, ResNet performed the worst. Since OCT images contain large irrelevant information, there is a lack of attention mechanisms to assist the network in identifying key feature areas. The F1 value of TCAM-ResNet was the closest to the results of the method in this paper. The TCAM-ResNet proposed a three-dimensional convolutional attention module (TCAM) that can extract weighted features from 3D images. The ACC values of these methods were relatively close, because the TF values in multi-classification tasks were high, resulting in generally high ACC values.

5. Conclusion

This paper proposes the Multi-step Interaction Network to unify processing of projection and feature fusion based on OCT and OCTA images. First, we introduce the Multi-modal Interactive Projection Module, which captures feature information at different resolutions using

a multi-scale projection approach. Subsequently, image and semantic information from both modalities interacts through shallow feature interaction operations. Next, we propose a Feature Redundancy Removal Module that enables the network to focus more on discriminative features when processing the differential feature information from the two modalities. Finally, we design a Feature Interaction Fusion Module that uses the attention mechanism to learn the interdependence between the feature information of each modality and the differential feature information, thereby enhancing feature through interaction fusion. In this article, we conducted ablation experiments and comparison experiments with other state-of-the-art and classical classification networks. The results demonstrate the excellent performance of multi-level differential and interactive fusion framework of MINet for multi-class classification of ophthalmic diseases based on 3D OCT and OCTA images.

However, there are very few publicly available 3D OCT and OCTA image classification datasets, so we tested the performance of MINet on

the OCTA-500 dataset, and the OCTA-500 dataset has a limited sample size.

The multi-step interaction framework presented in this paper can be extended to other similar multi-modal 3D image tasks, such as Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) images, in the future.

CRedit authorship contribution statement

Han Zhang: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Xuening Bai:** Writing – original draft, Validation, Software, Methodology, Formal analysis. **Guangyao Hou:** Formal analysis. **Xiongwen Quan:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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